

Swedish Hypersonics from Hermes on

J.Oppelstrup, KTH (em.)

Focal areas

Aerothermodynamics: Shock layers, TPS

Propulsion: PDE and Scramjets

(Materials) ..

Players

FOI: Swedish Defence Research Agency
--2000: Mainly experimental
Hypersonic tunnels HYPxxx
2000 – Computation / test



SAAB:
CFD



KTH:
Real gas CFD
Materials



LTH:
Combustion



GKN Aerospace:
Rocket nozzles



Küchemann book – DynaSoar, wave riders

Swedish hypersonics

1. Hermes era -- 1992

Shock-vortex interaction on blunt deltas
TPS heating
Equilibrium real-gas CFD.

2. The quiet period 1992 -- 2010

FOI missile aerodynamics
Shock-layer modelling
PDE model and experiments

3. Modern FOI & LTH hypersonics

Shock-layer/vortex coupling on gliders
High-enthalpy modelling
Scramjet inlet physics.
LES/DNS shock–turbulence interaction
reacting flows

4. HYPER-X initiative 2026 --

Modern tools, unified collaborative agenda.

Sweden in Space: Esrange Space Center

*“Once the rockets are up, who cares
how [where] they come down,...”* (Tom
Lehrer)

Small payloads to orbit
Satellite Comm.(68 ° N)
Rocket testbed



Esrange shelters:
56,000 km² Sami country



Hermes

Blunt Delta Wing $M \leq 7$ WT tests

Equilibrium real gas in Saab CFD

Jan Vos, then EPFL

Academic work on numerics

γ - gas

Shock capture

Boundary layer BC

Focus on delta wing

Shock-Vortex-BL interaction

--- dual purpose ---

Saab 39

VFE

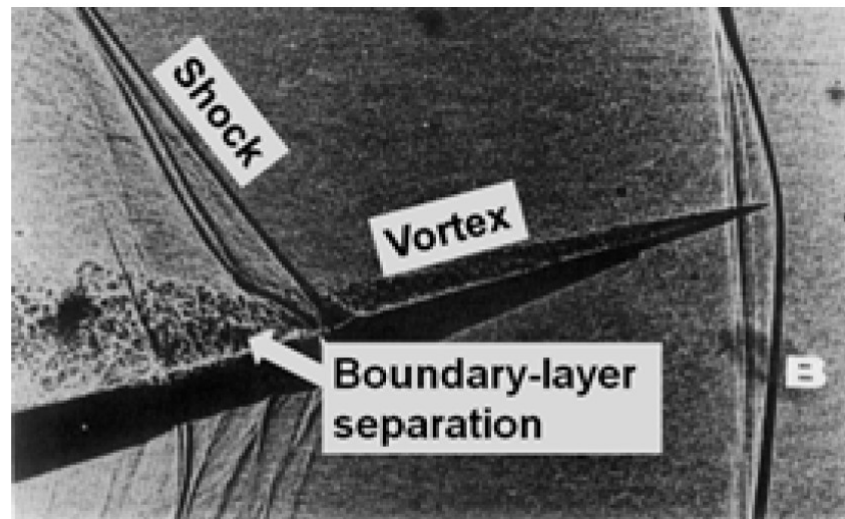


Figure 2.24 Shock-Vortex-Boundary layer interaction at supersonic speed. (Unpublished, courtesy of Anders Gustafson, Swedish Aeronautics Research Establishment FFA.)

FOI-R--0135--SE (2001)

Multi cycle experiments on a pulse detonation engine

Tegner,J., Olsson,S.

PDE: PULSE DETONATION ENGINE



FOI-R-4991-SE (2020)

Aerodynamic Modelling of Hypersonic Glide Vehicles

Nordin,J., Hedlund,H., Karlsson, M

FOI-R-5143-SE (2022)

High-Speed Aerothermodynamics for Advanced Flight Vehicles

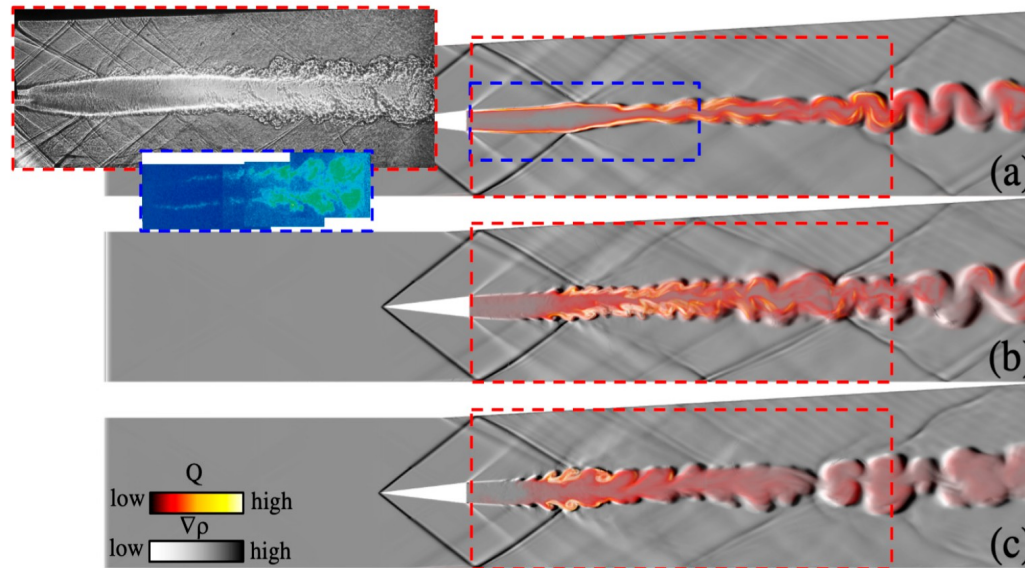
Karlsson,M., Hedlund,H.

CATLA Clean Air High Temperature Laboratory (2023)

Experimental facility, scramjet

1700 K clean, dry air at ~25 km altitude conditions





Density gradient, $\nabla \rho$, combined with the heat release, Q , for the (a) D7, (b) J20, and (c) Z22 reaction mechanisms using the LDKM subgrid flow model



LUND
UNIVERSITY

HYPER-X

Hypersonic sYstems for Pioneering Excellence and Research eXchange

C. Fureby et al



Purpose: Develop *knowledge* and *key enabling technologies* and *methods* for hypersonic flight and address societal implications.

Vision: The vision of HYPER-X is to establish Sweden as an internationally renowned partner in hypersonic systems research and innovation by 2035 including proof-of-concept testing.

- Revolutionize civil aviation and space access through reusable hypersonic vehicles.
- Supports climate goals using H₂-based propulsion for net-zero carbon aviation.
- Advances defense capabilities and strengthens European security.

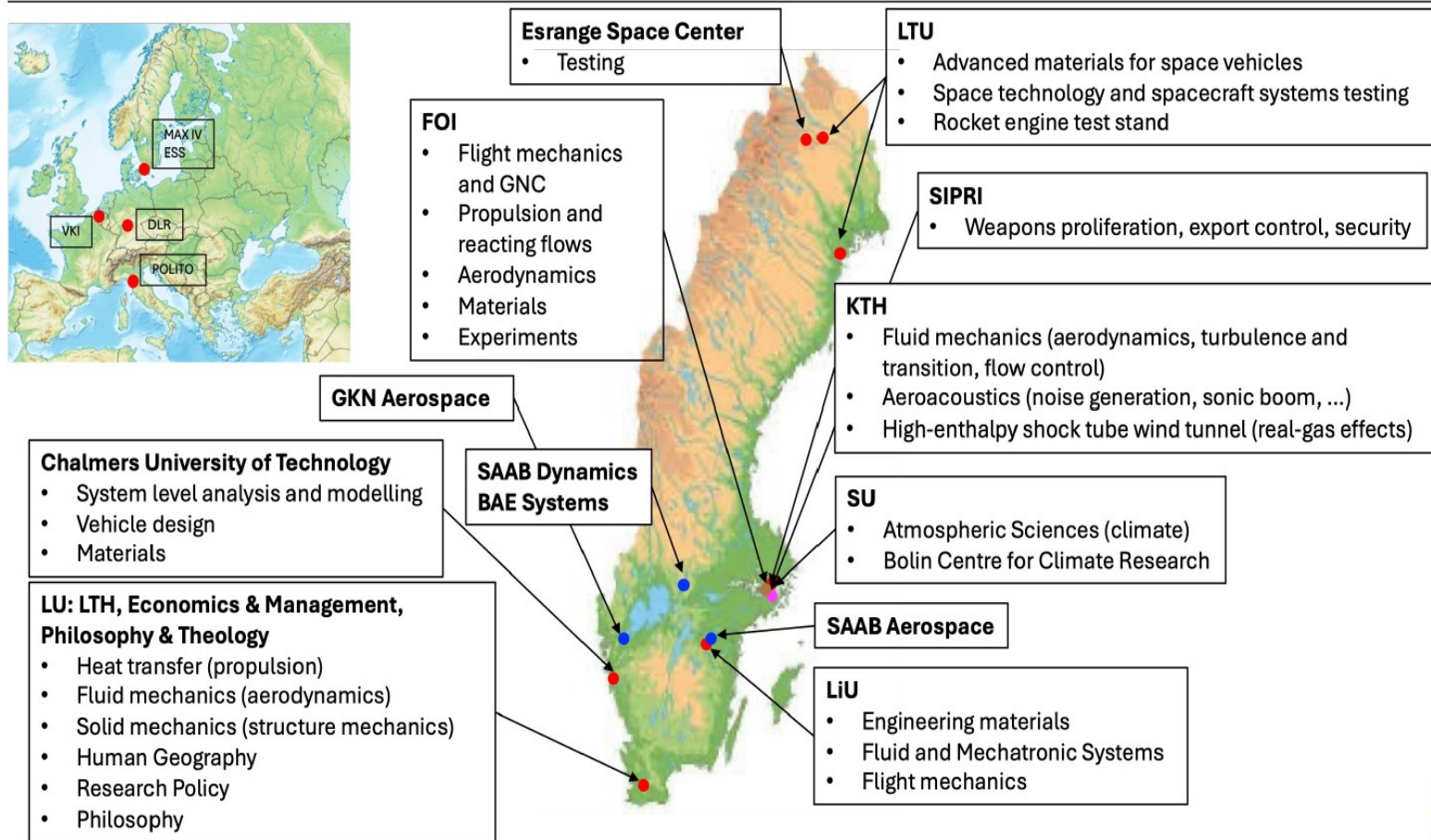
Scope: Research in aerodynamics, propulsion, thermal protection, structures, materials, air vehicle design, flight mechanics, guidance, navigation and control, manufacturing, climate, social acceptance, proliferation, techno-economy, ...

Benefits: Technological sovereignty, industrial innovation, sustainable aviation, economic growth, global competitiveness, bringing the world closer together.

Workshop: Held in Jan 2026 to prepare for an application



HYPER-X Team



Summary

After a lengthy period of work-up exercises following the demise of Hermes:

*The HYPER-X project can become
a competent trusted partner in
collaborative projects*